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AI for Bharat Hackathon

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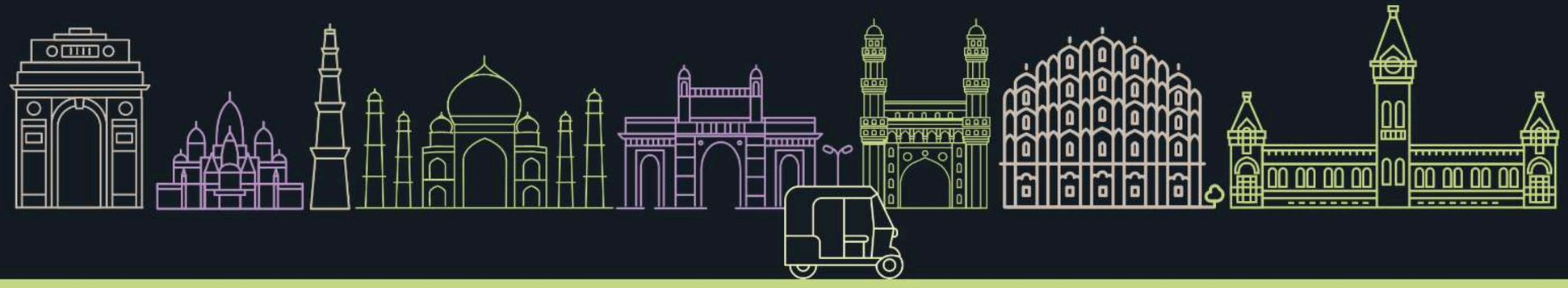


Team Name : OOPs We Coded Again

Team Leader Name : Faheemuddin Sayyed

Problem Statement : Build an offline-first, multilingual AI tutoring system that delivers curriculum-aligned education to rural students despite limited internet access, teacher shortages, and language barriers (AI for Communities, Access & Public Impact).

Brief about the Idea:

THE CHALLENGE: A DEEP DIVIDE IN RURAL EDUCATION

1 MILLION+

Teacher Vacancies Nationwide

Rural elementary schools face a net deficit of more than 286,000 teachers.

LESS THAN 4%

High-Speed Internet Access

Only 3.8% of rural areas have fibre access, with household access as low as 24%.

OVER 104,000

Single-Teacher Schools

This reflects insufficient resource allocation and affects the quality of education available.

THE SOLUTION: DIGIMASTERJI

Fully Offline Capable

A Voice-First, Offline AI Tutoring System

Designed to function in low or no-connectivity environments on shared devices.

Overcomes Language Barriers with Multilingual Support

Offers voice-based tutoring in regional native languages to improve inclusivity.

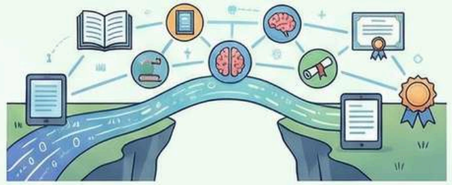
Curriculum-Aligned for Scalable Learning

Ensures syllabus compliance while providing accurate and personalised education.

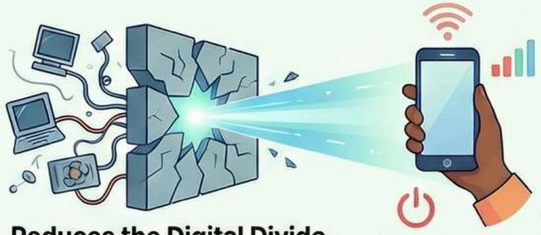
Benefits And Impacts

Social Impact

Fosters Educational Equity
Democratizes high-quality STEM education for millions of students in rural areas.



Bridges Learning Gaps
Fills educational voids from teacher shortages with personalized, adaptive learning.



Reduces the Digital Divide
Makes advanced AI accessible without reliable internet or expensive devices.

Night School: The "Audio-Only" Advantage

Reduces Strain
Offers a dedicated Audio-Only Mode for late-night study or exhausted students.

Increases Accessibility
Allows users to listen and learn while resting, commuting, or in low-light conditions.

Promotes Inclusive Learning
Supports students with visual impairments or those with limited screen time.

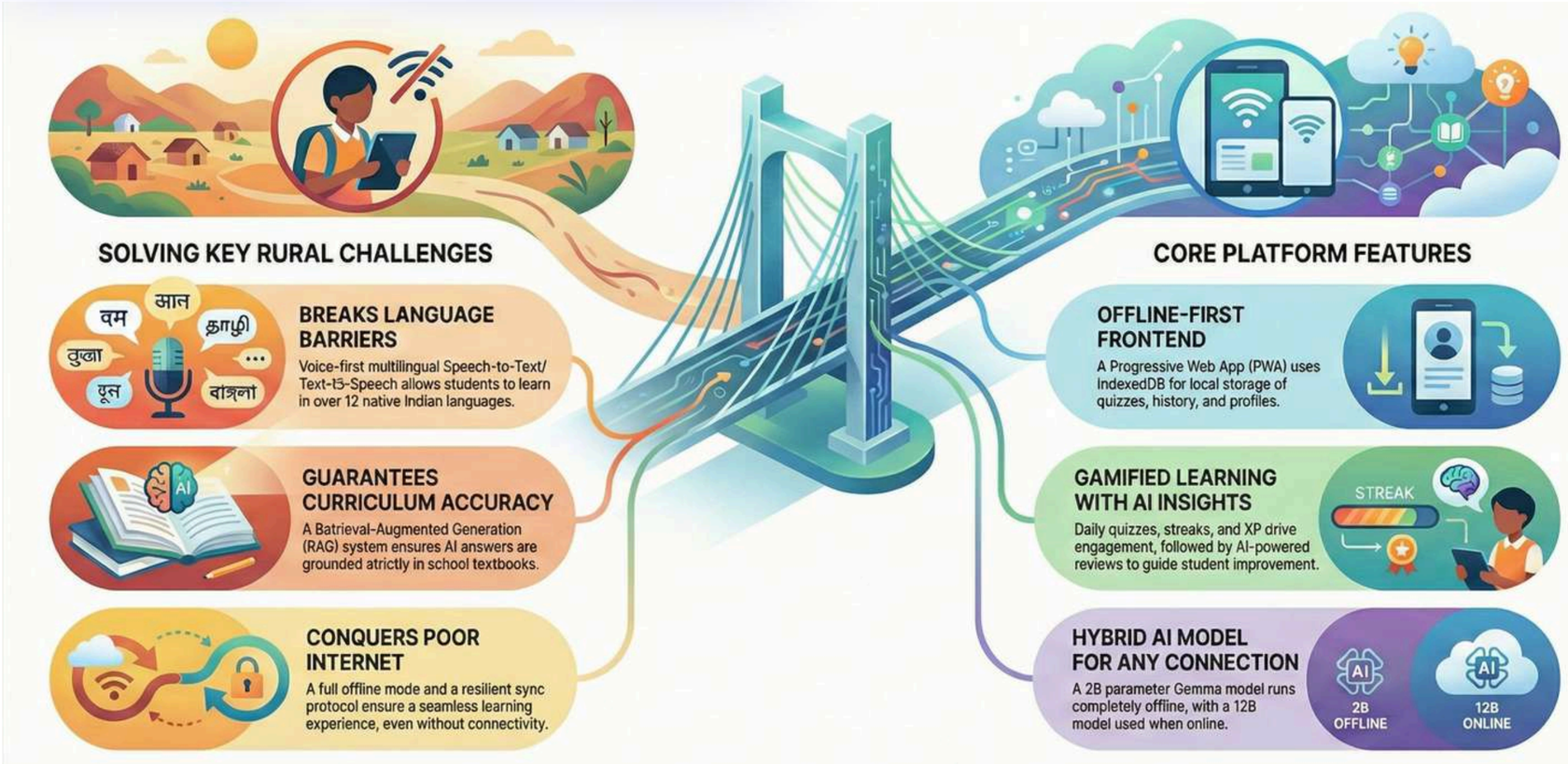
Comparision

Feature	Generic AI Chatbots	DigiMasterJi
Curriculum Grounding	Broad, internet-based (can hallucinate)	Strictly Textbook-Aware RAG: Syllabus-compliant answers.
Offline Access	Requires constant internet	Full Offline Functionality: Learn without internet.
Multilingual Audio	Limited/API-dependent	Voice-enabled (12+ Indian Languages): Whisper + TTS.
Cost Per User	API fees scale rapidly	Open-Source/Local AI: Highly cost-efficient at scale.
Rural Focus	General-purpose	Designed for Rural: Addresses connectivity, language, cost.

Solution Breakdown

Key Innovations and Unique Differentiators

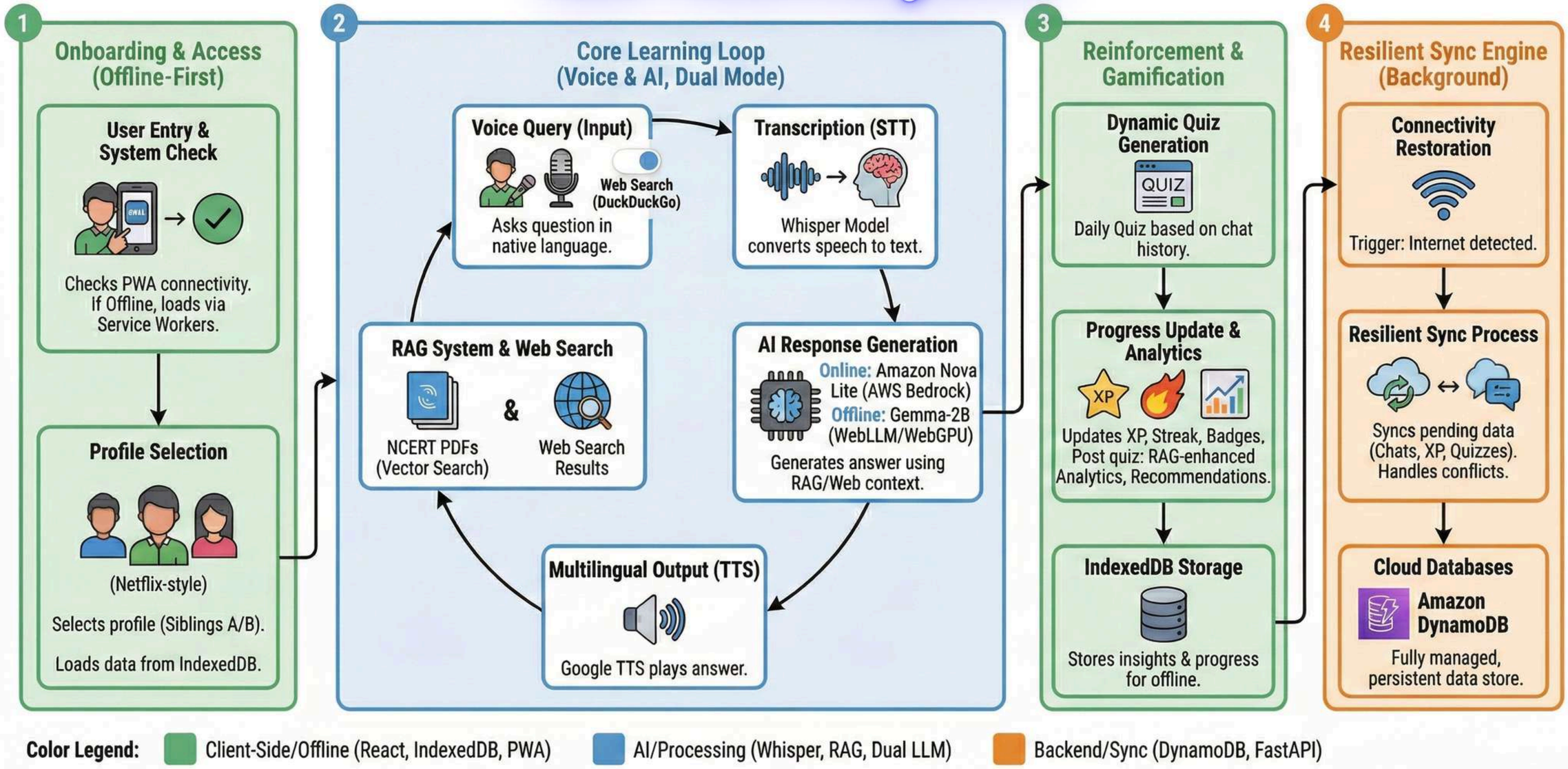
- **Transient Privacy:** Audio is processed for transcription then instantly deleted to ensure 100% student data security.
- **Multi-Student Profiles:** One device, independent learning paths; unique progress tracking for multiple siblings.
- **Curriculum-Aware RAG:** Not a chatbot—a dedicated tutor grounded strictly in your school's PDF syllabus.
- **Multi-Day Sync:** Custom conflict-resolution designed for students who remain offline for days at a time.



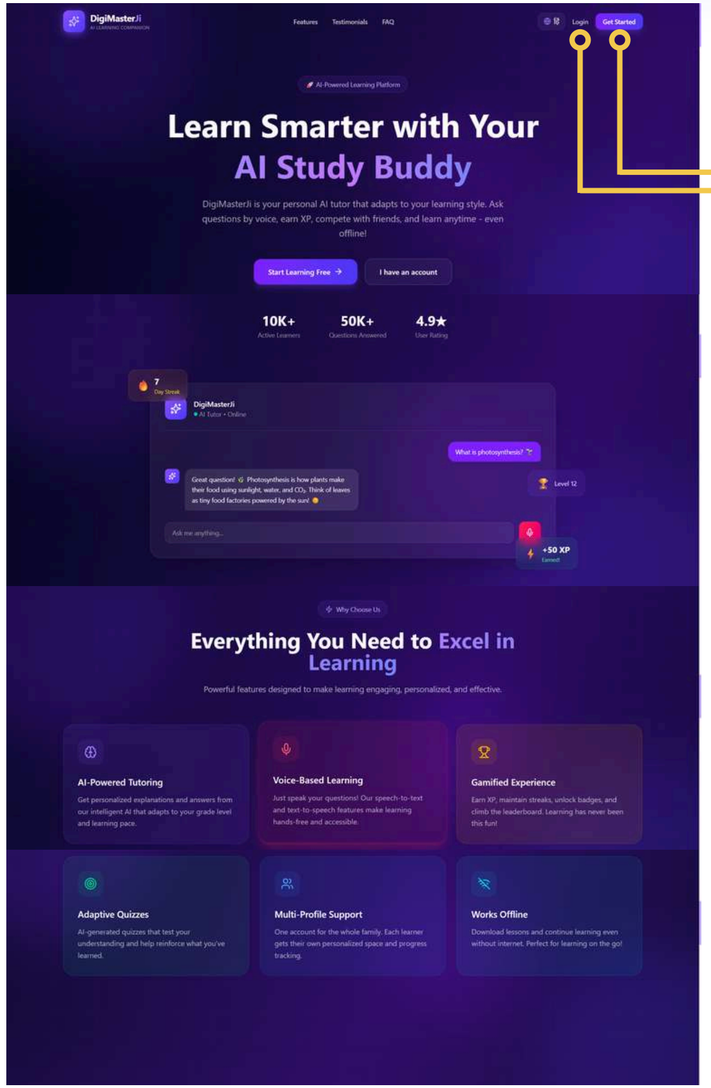
How the Solution Directly Solves the Identified Problem

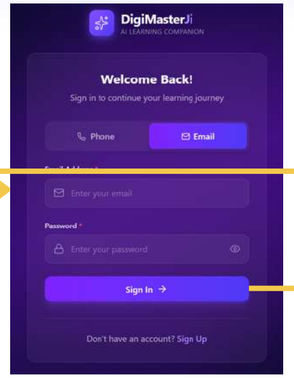
- **Poor Internet → Offline Sync Protocol Full PWA functionality (Quizzes/Chat) via cached models; background sync only when online.**
- **AI Hallucinations → Curriculum RAG System Knowledge grounding restricts AI to verified textbooks, ensuring 100% syllabus accuracy.**
- **Language Barriers → Voice-First Vernacular Multilingual STT/TTS allows students to learn STEM in native languages without high literacy barriers.**

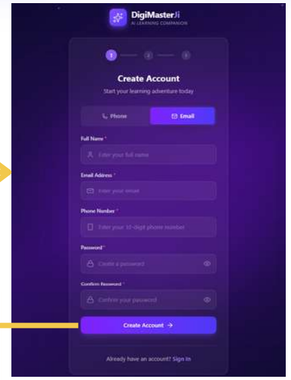
Use-case diagram

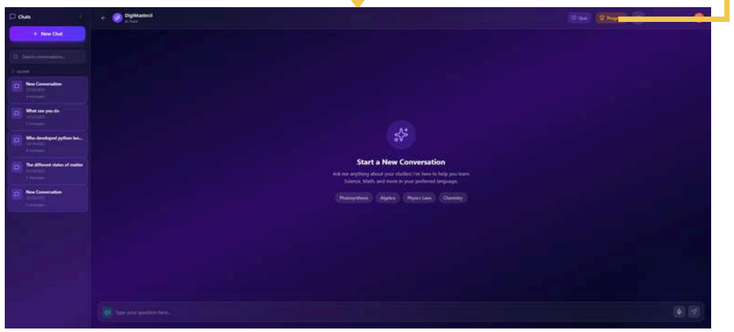


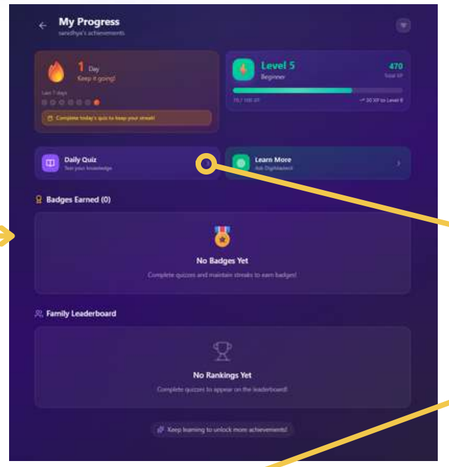
Wireframes/Mock diagrams

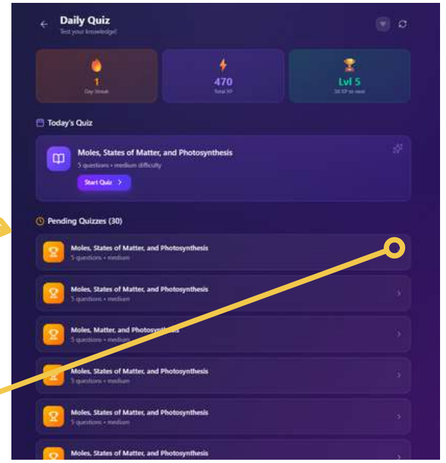


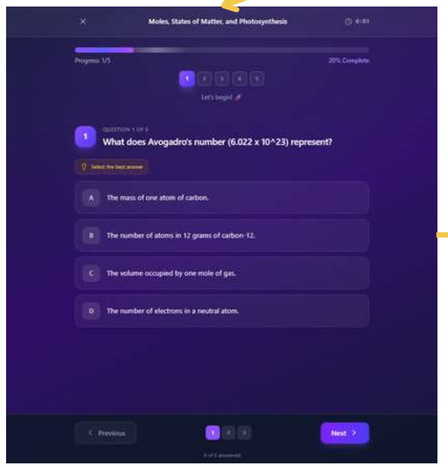


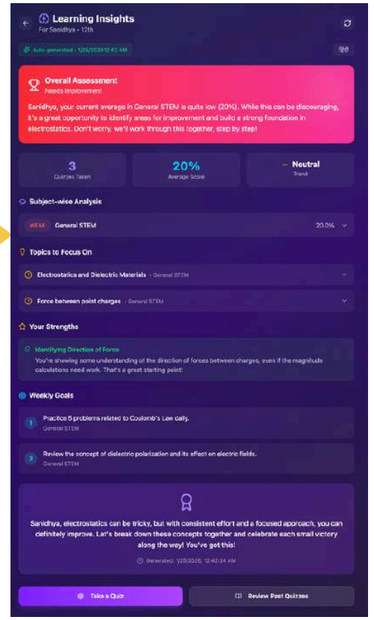




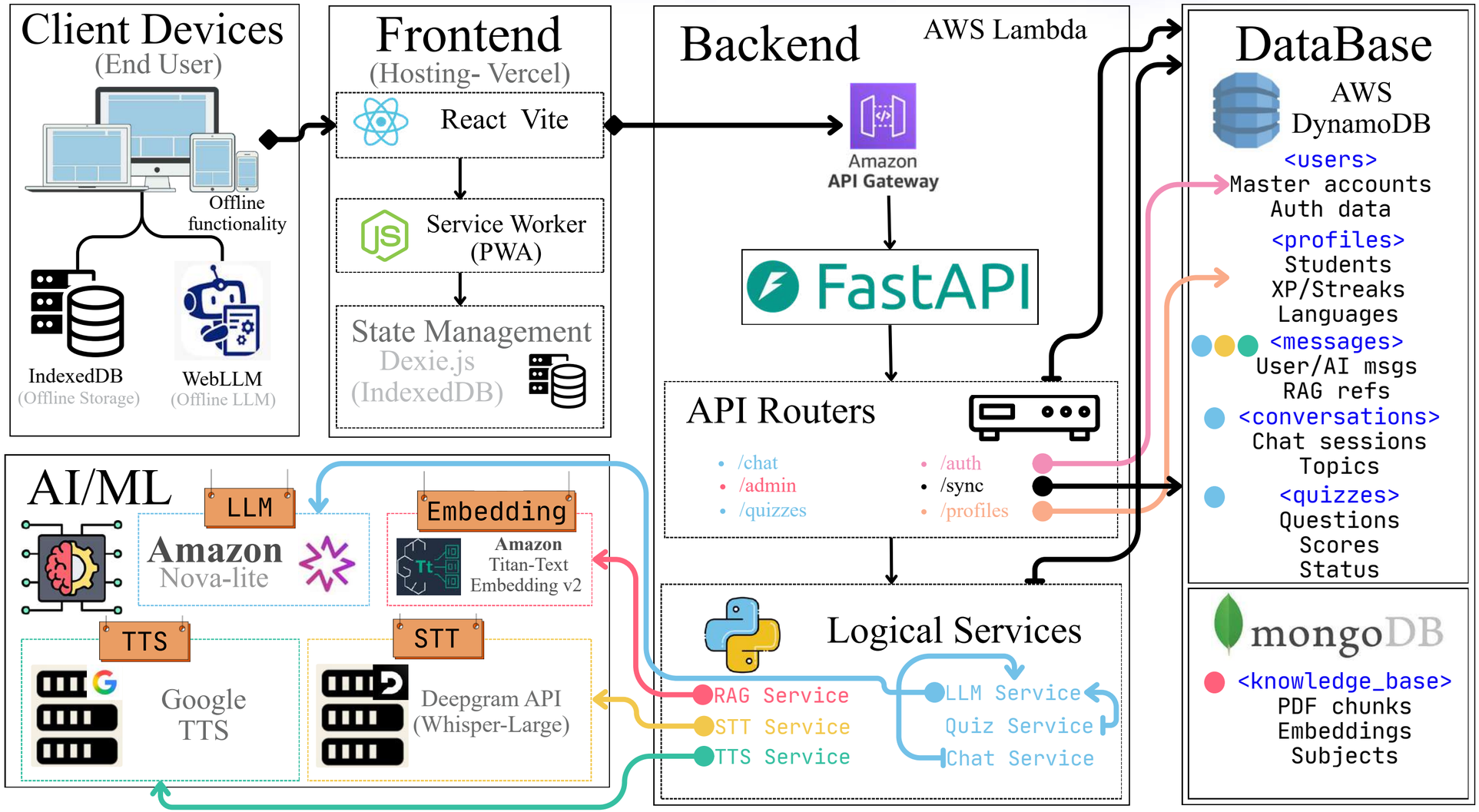








Architecture diagram:

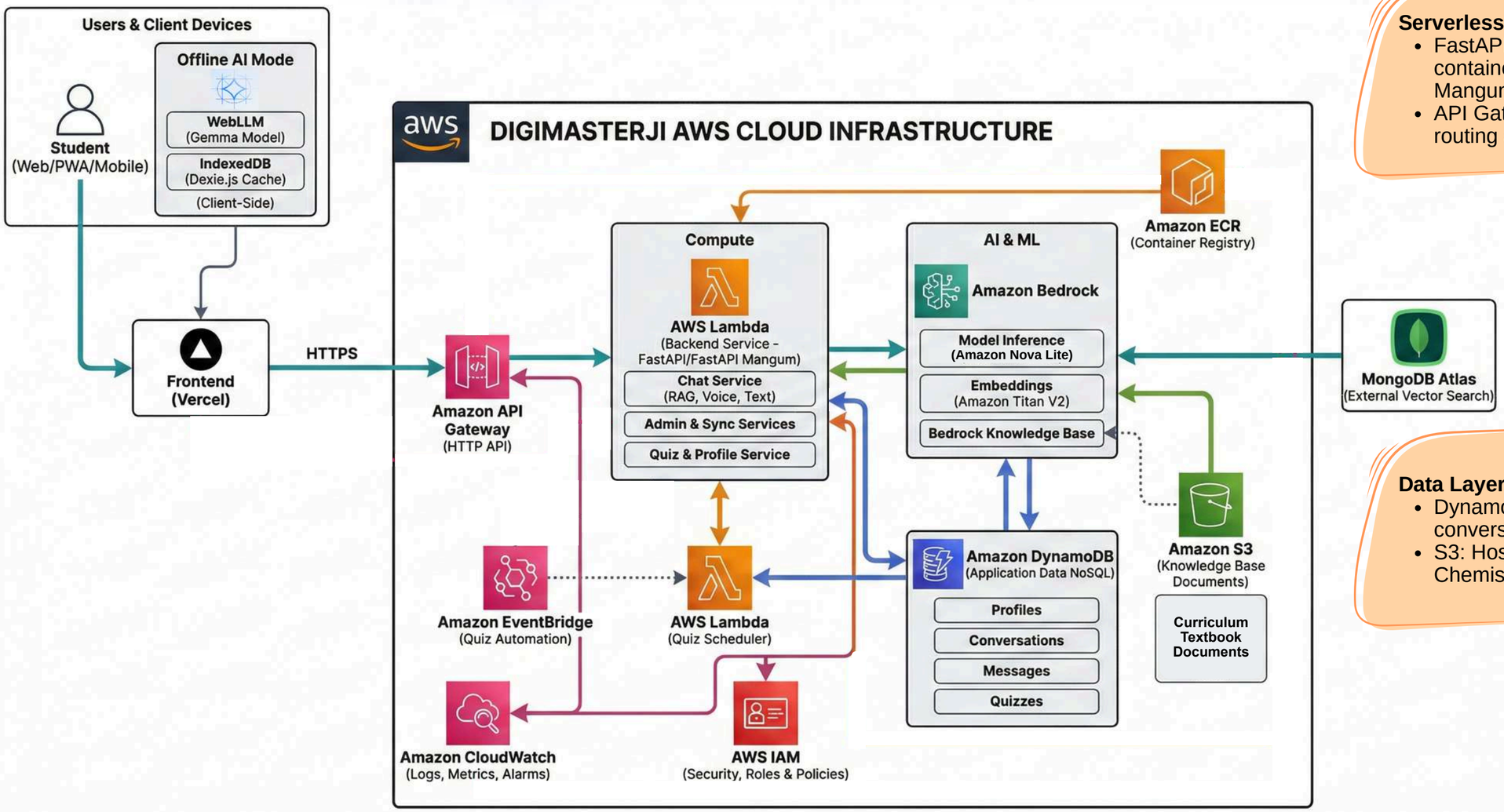


WebLLM Offline AI:
True offline capability with Gemma 2B model running entirely in-browser via WebGPU.

Quiz automatically generated by EventBridge daily for each profile based on chat history and profile data

Curriculum Textbooks are fed into the RAG Vector Database Using Admin Dashboard

Deployment Cloud Architecture diagram:



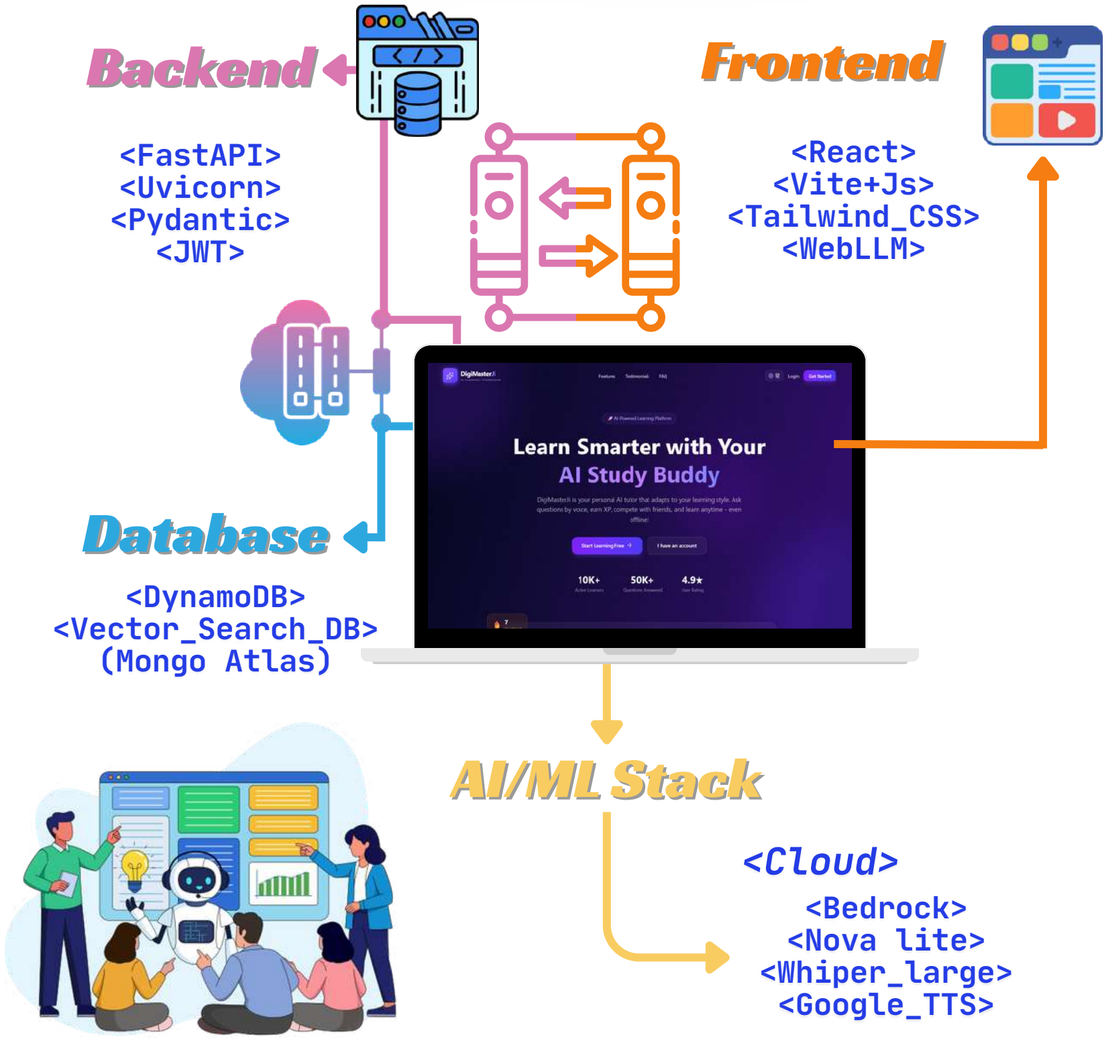
Serverless Compute
- FastAPI backend deployed as container on AWS Lambda via Mangum adapter
- API Gateway (HTTP API) handles routing and HTTPS termination

AI/ML Pipeline
- Amazon Bedrock: Nova Lite for LLM inference, Titan V2 for embeddings
- Bedrock Knowledge Base with MongoDB Atlas as vector store for RAG

Data Layer
- DynamoDB: Stores profiles, conversations, messages, quizzes
- S3: Hosts curriculum PDFs (Physics, Chemistry, Math, Biology)

Automation & Ops
- EventBridge: Schedules daily quiz generation per student
- CloudWatch: Centralized logging, metrics, and alarms

Tech-Stack



- **PWA Frontend (React + Vite):** Lightweight, fast-loading Progressive Web App designed for low-end devices and unreliable rural connectivity, with full offline support.
- **FastAPI + AWS Lambda Backend:** High-performance, serverless APIs with Mangum ASGI adapter enabling auto-scaling, pay-per-use compute, and seamless deployment on AWS .
- **DynamoDB + MongoDB Atlas:** DynamoDB for fast NoSQL storage (users, profiles, chats, quizzes); MongoDB Atlas for vector search enabling curriculum-grounded RAG within free-tier limits..
- **Amazon Bedrock (Nova Lite + Titan V2):** AWS-managed LLM for multilingual STEM tutoring with Titan Embeddings V2 for semantic search—serverless AI with no GPU infrastructure required.
- **Multilingual Voice (Deepgram + gTTS):** Cloud-based Whisper STT via Deepgram for 15+ Indian languages; Google TTS enables cost-free voice output in Hindi, Tamil, Telugu & more.

Estimated implementation cost:

The Prototype Phase (5–10 Users)

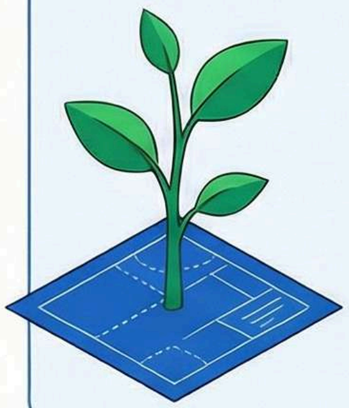
Total Cost: ~$1.47 per month

Minimal expenses for a small developer team and test group.

The Power of Free Tiers

Lambda | API Gateway | MongoDB Atlas M0

Lambda, API Gateway, and **MongoDB Atlas M0** tiers result in **£0.00** charges.



AI/ML is the Primary Expense

Amazon Bedrock

Amazon Bedrock accounts for **over 50%** of the initial prototype budget.

Comparison of Top Cost Categories

Early Product Release (500–1,000 Users)

Total Cost: ~$85.30 per month

A significant jump as usage exceeds free tier limitations.

Compute and Monitoring Spikes

Lambda and **CloudWatch** become the dominant infrastructure costs.

Lambda ($25.40)

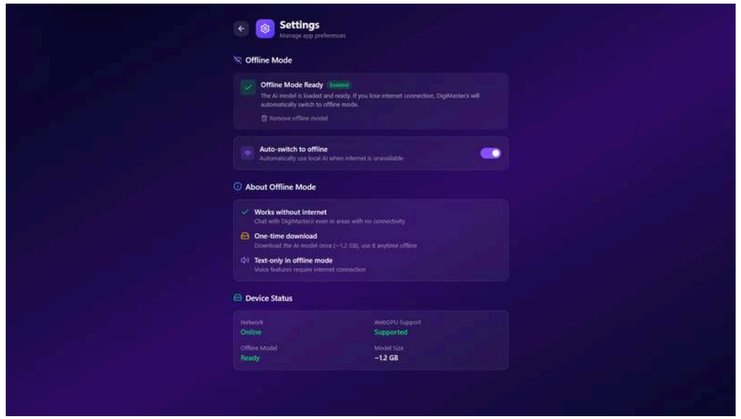
CloudWatch ($20.60)

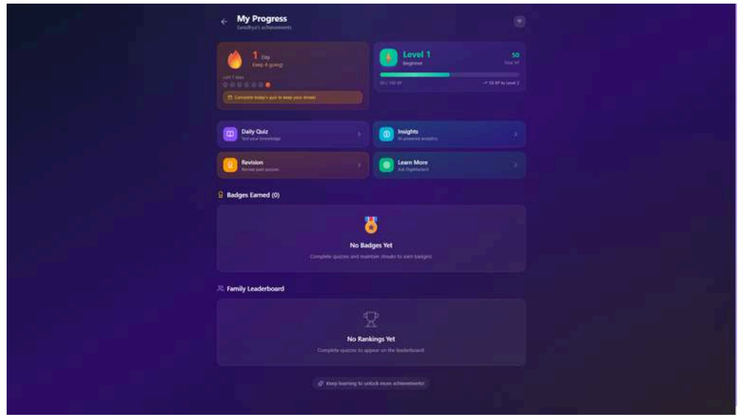
Database and AI Growth

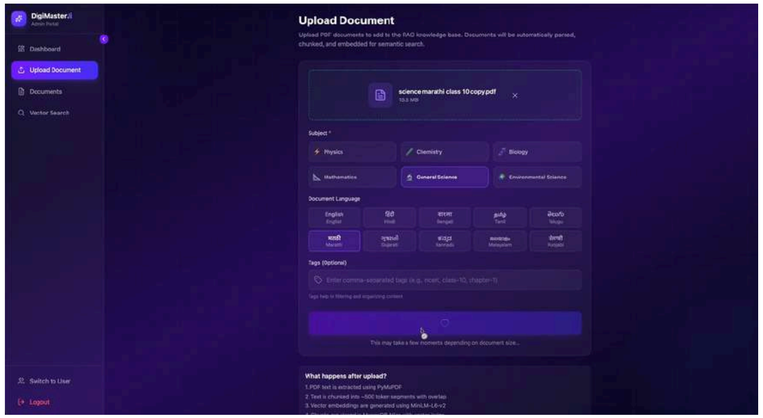
Bedrock AI costs scale to **$16.00**

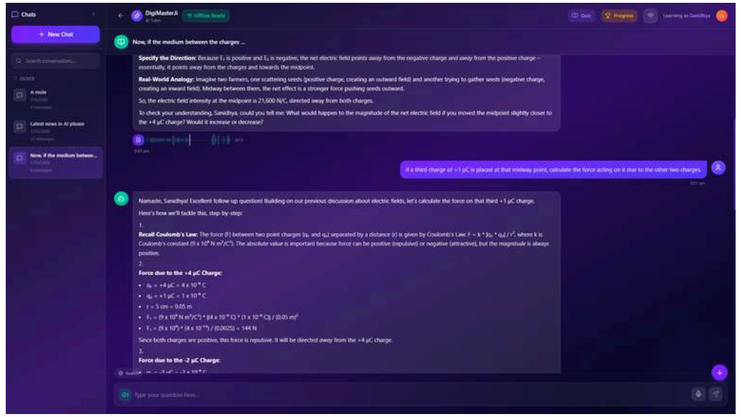
MongoDB shifts to a paid tier

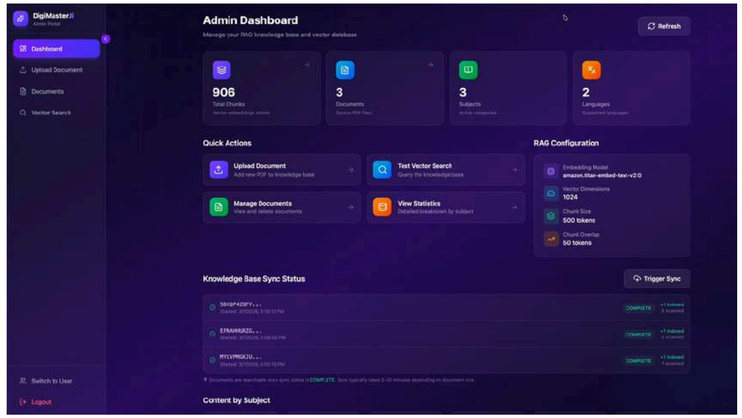
Snapshots of the prototype:

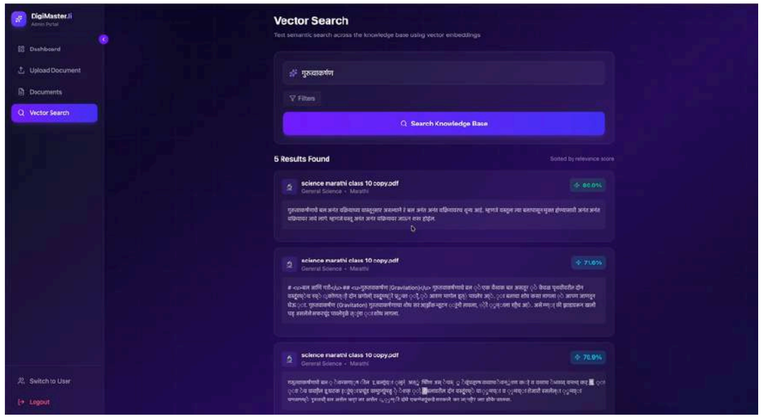












Prototype Performance report/Benchmarking:

AI MODEL LATENCY & ACCURACY

Offline Inference (Gemma 2B)

Avg Response Time: **5-7 seconds** via WebGPU/WebLLM on mid-range devices

Cloud Inference (Nova via Bedrock)

Total Response Latency: **~1s** (Avg 914ms), sub-second TTFT

RAG ACCURACY & CURRICULUM INTEGRITY

Retrieval Precision

95%+ Verified against NCERT Physics & Biology textbooks

Hallucination Control

Ensures curriculum-based, hallucination-free queries

SYSTEM EFFICIENCY & FOOTPRINT

Cold Start Optimization

<150ms Warm Start
Optimized using FastAPI Mangum (Avg 134ms)

Offline Sync Throughput

50+ Cached Interactions
Synchronized in single 5-second internet burst

Storage Footprint

PWA Size **<15MB** (Excl. cached model weights)

VOICE PROCESSING & SCALABILITY

Transcription Speed (Whisper via Deepgram)

10s Audio processing:
Hindi <5s, English <3s

TTS Naturalness

gTTS Low-Latency Audio
Supports Hindi, Tamil, Telugu

Simulated Scalability

200+ Requests/sec
Handled by AWS Lambda/API Gateway

Database Performance

DynamoDB: **<10ms**
Read/write latency for student profiles

Metrics verified through benchmarking code in `**Backend/benchmarks**` of the GitHub repository.

Additional Details/Future Development (if any):

Evolving into a robust offline learning ecosystem, optimizing AI for mid-range devices and introducing advanced pedagogical tools.

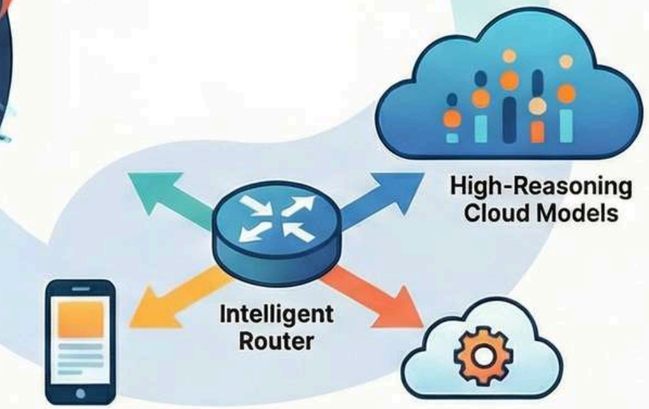
Technical Infrastructure & Optimisation

High-Performance Offline AI

4-bit and 8-bit quantization

Local Edge Computing

Using village school servers as local caches to reduce individual device download requirements.



High-Reasoning Cloud Models

Intelligent Router

Cheep Local Models

Smart Hybrid Routing

Automatically routing queries between local and cloud-based models.

OCR for Handwritten Work

Enabling students to photograph physical notebooks for automated AI transcription and grading.

Enhanced Learning & Accessibility

Regional Dialect Scaling

Fine-tuning voice models to recognise specific regional accents within major Indian languages.

Stakeholder Dashboards

Providing multilingual SMS summaries for parents to track student progress and learning gaps.

Prototype Assets:

- GitHub Public Repository Link :
  - https://github.com/FaheemAlt/DigiMasterJi
- Demo Video Link :
  - https://youtu.be/h_Y3sgLOPy8
- Live Prototype :
  - https://digimaster-ji.vercel.app/

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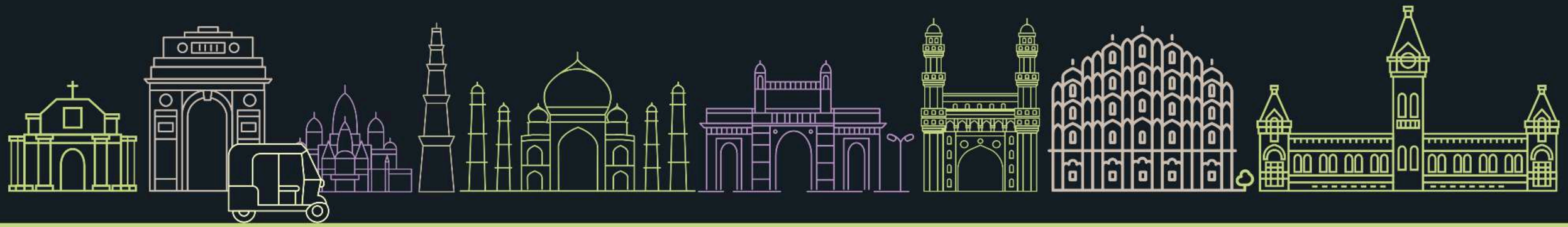
Thank You